# Peter Morganelli

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## Education

**Tufts University** — Medford, MA
*Bachelor of Science in Computer Science* — *Expected May 2027*

- **Dean's List (6x)**
- **Coursework:** Algorithms, Data Structures, Concurrent Systems, Networks, Natural Language Processing, Machine Structure & Assembly, Intro Cybersecurity

**Trinity College Dublin** — Dublin, Ireland
*Semester Abroad, Computer Science* — *January 2026 – May 2026*

## Technical Skills

**Languages**: C++, C, C#, Python, JavaScript, TypeScript, SQL, x86 Assembly
**Frameworks & Libraries**: React, React Native, Next.js, Node.js, Django, Spring Boot, Three.js, Unity
**Infrastructure & Tools**: Docker, AWS (S3, EC2), Vercel, PostgreSQL, MongoDB, Neo4j, Redis, Firebase, Vitest, Git

## Experience

**Software Engineer Intern** | *Python, Django, Neo4j, Docker, GitLab* — Bedford, MA
*MITRE* — *May 2026 – Present*

- **Parallelized** bounded requirement verification for a **GraphRAG** application with a a **four-worker** async request pool with built-in **retry** and **rate-limit** handling, cutting end-to-end runtime by **77%** on large Excel imports
- Shipped production features on a **Django** requirements platform over a **Neo4j** graph database, maintaining traceability between source requirements, system-model data, and LLM-generated findings for Air Force **CUI** systems
- Presented **AI agent orchestration** and model prompting workflows to **50+ senior engineers**
- Fixed **90+** SonarQube issues, code refactors, security hotspots, and code smells for compliant **DoW** standards

**Data Structures Teaching Fellow** | *C++, GitLab, Gradescope* — Medford, MA
*Tufts University* — *September 2024 – Present*

- Own grading standards for a **200+ student** course, setting the staff-wide call on ambiguous rubric criteria across **300+ submissions**
- Revise project specifications and lecture material with faculty; support students in office hours and on Piazza

**Full-Stack Developer Intern** | *React, JavaScript, PostgreSQL, MongoDB, AWS* — Medford, MA
*Launch* — *February 2025 – April 2025*

- Contributed to an **AI-powered website builder** from scratch at an early-stage startup, working across the **React** frontend and the **RAG** pipeline
- Designed a **drag-and-drop editor** for end users to restructure and customize generated site components

**Software Developer** | *React, TypeScript, TailwindCSS, Figma* — Medford, MA
*Tufts JumboCode* — *October 2023 – May 2025*

- Built a **full-stack** inventory site for a Somerville theater with **10+ developers**, implementing **RESTful API** handlers and **unit tests** across 10+ components
- Shipped an image-upload carousel with per-item metadata and filter tags from a **Figma** prototype

## Projects

**Peter's Pizzeria** | *React, Vite, Vercel Serverless, Vercel Blob, Upstash Redis, Vitest* — May 2026 – Present

- **Launched** a full-stack ordering platform for my student-run pizzeria with customer checkout, a live kitchen board with **5s polling** configurable store hours, and community image posting across **15 serverless endpoints**
- Hardened the order API against client tampering by recomputing every price **server-side**, authenticating staff with **HMAC-derived tokens** in HttpOnly cookies, and enforcing Redis-backed per-IP **rate limits**
- Wrote a **140-test Vitest suite** that runs on real HTTP handlers, then checked it via mutation testing for enforcement of accurate unit testing

**Programming Languages Interpreter** | *Elm, JavaScript, HTML/CSS* — May 2025 – August 2025

- Engineered a zero-setup **web-based IDE** integrating five language interpreters via **Elm ports** and JavaScript bridges for live in-browser execution
- Designed a **modular interpreter API** for adding new languages, built for **200+ students** to complete coursework, under the mentorship of **Professor Emeritus Dr. Norman Ramsey**